Oncology Analysis

2026-04-23

Quality Control

The use of PCA is used to reduce the dimensions of the data. Instead of the data being in multiple dimensions, several variables, PCA compresses the data down to principal components. This finds the directionality of the data and allows for the data to be compressed into the important components. There is a difference between scaled and unscaled, when using scales, PCA effectively uses the covariance matrix for unscaled and the correlation matrix for scaled. This means that using unscaled gives large scales to dominate the sensitivity and magnitude. It does mean that the importance of features is based on the true variance. For scaling, the data variables are forced into a mean of 0 and a variance of 1 before PCA. If values are measured on different scales, this has no impact. The issue is that the variables are treated equally, which means that a variable that is more important than another is still treated the same. In our analysis, using PCA on unscaled data worked best, as we were looking at cancer type and how it related to an individual; it created clear clusters of where the data was concentrated. Clear clustering of low-grade gliomas, medulloblastoma, ganglioglioma, both Ependymal tumours and High Grade Gliomas overlap considerable. What this likely means is that low-grade gliomas have a protein that is consistently expressed across these patients, such as BRAF V600. There is internal heterogeneity, that multiple there a multiple subtypes under one label. For medulloblastoma and ganglioglioma, there is a protein which dominates.

PLS-DA scores which was run on prot_imputed as the X and cancer detailed type as the Y:

Supervised metric unlike PCA.

Medulloblastoma separated, confirms distinctive proteome

Ganglioglioma, separate well, very consistent with PCA

Low-grade glioma, confirms heterogeneity, as forces of separation, it still remains sprawled

Craniopharyngioma and ependymal overlap a lot, suggesting proteomes are similar

No clear elbow plot flattens after component 3, 33% to 25%. Using 5 maximises performance. It suggests using the balanced error rate, that the classes are imbalanced, so finding clusters for the model is easier.

Contribution 1 plot: The colour tells you which subtype each protein is most associated with here, almost everything is pink (Medulloblastoma), or black (PLGG), which makes sense since component 1 primarily separates those two large groups.

EHMT2, SMC3, MAD2L1, MSH2, PCNA, MLH1 all point towards Medulloblastoma. These are DNA repair, cell cycle, and chromatin regulation proteins, consistent with the high proliferative activity of medulloblastoma

PRKCA, SEPT8, RHOG — pointing towards PLGG (negative direction). PRKCA (Protein Kinase C alpha) is interesting given BRAF pathway activity in many PLGGs

Contribution 2 plot: Ganglioglioma-associated proteins (green, positive direction): KIF5C, HOMER1, KALRN, WASF1, YWHAG, GPRIN1, GPHN, CLVS1, BRSK2

Craniopharyngioma-associated proteins (orange, negative direction): DUSP14, LRRC1, CASP8

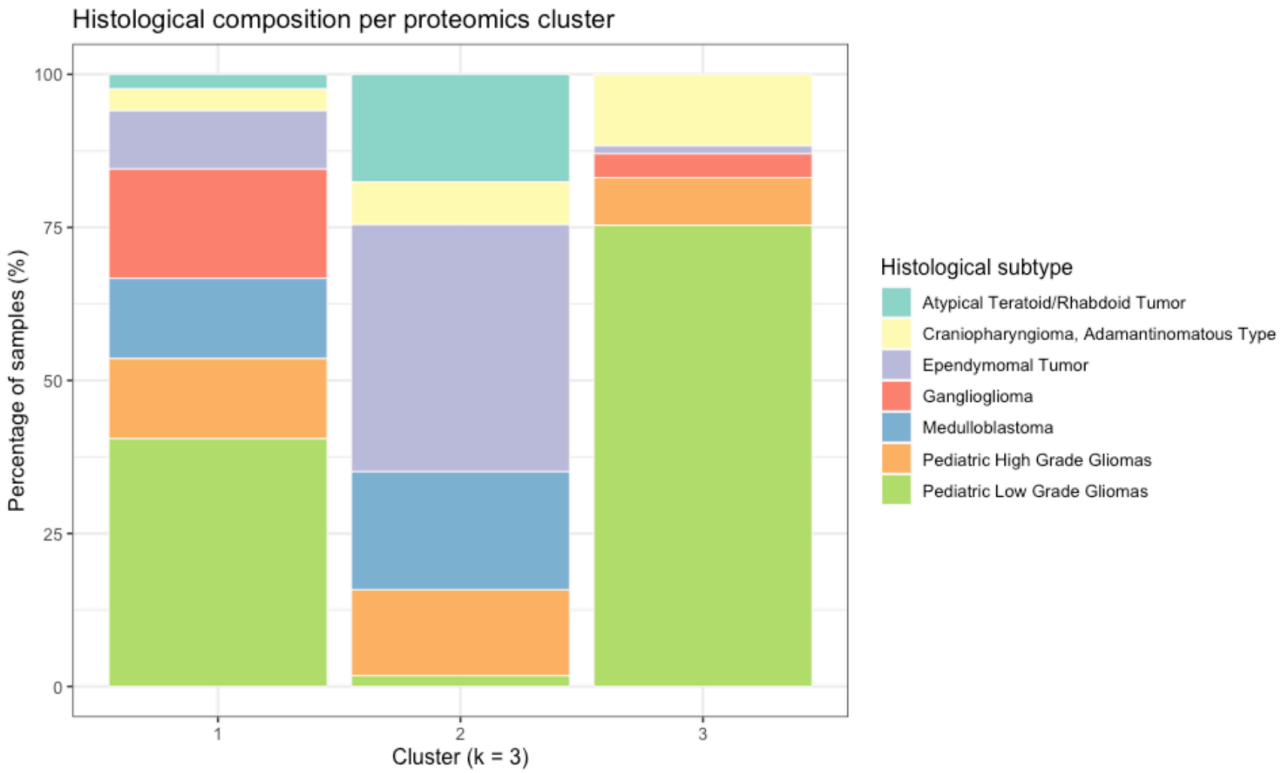
Ependymal: ERBB2

The next step was looking at cluster analysis. First, using NMF, Non-Negative Matrix Factorisation. A logging process to test different numbers of factors and find their rankings to find the optimal rank. There was no clear result, but between 3 and 5 seemed like the logical step. Using the clusters rather than the variables means we can map what proteins have prognostic meaning, not an individual's proteins.

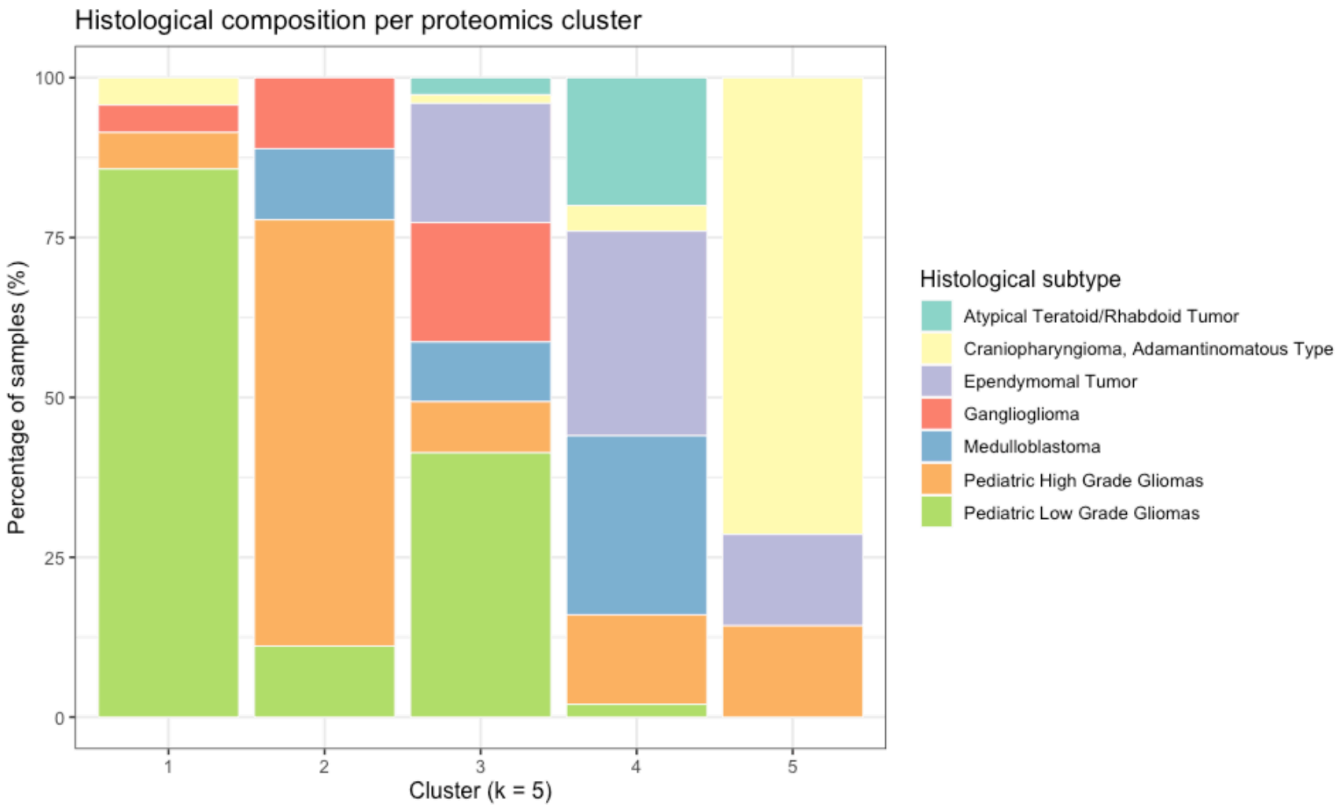
There are over 6000 proteins, and not each one can be placed into a model. This would cause an overfitting issue. The rule-of-thumb is for every 10-15 deaths or events, a variable is added. For this reason, using clusters of protein levels. The clusters act like a proxy of the underlying protein state.

Metric	3 clusters	5 clusters	Better
Concordance	0.715	0.754	5-cluster
LRT p-value	0.000253	5.89e-05	5-cluster
Score p-value	4.01e-05	9.59e-06	5-cluster
AIC	321.6	318.0	5-cluster
Significant clusters	1/2	2/4	5-cluster
Max HR	4.2 (C2)	6.2 (C4)	5-cluster

Cluster	n	Events	Mean OS (months)	Median OS (months)	HR	Dominant subtype	Other subtype
Cluster 1	84	15	47	34		Pediatric Low Grade Gliomas	Ganglioglioma, Medulloblastoma, Pediatric High Grade Gliomas
Cluster 2	57	24	51	39	1.439***	Ependymomal Tumor	Medulloblastoma, Atypical Teratoid/Rhabdoid Tumor
Cluster 3	77	8	48	36	-0.002	Pediatric Low Grade Gliomas	Craniopharyngioma, Adamantinomatous Type, Pediatric High Grade Gliomas
Sex	-	-		-	0.00	-	
Age	-	-		-	-0.178	-	



cluster	n	Events	Event Rate	Mean OS (months)	Median OS (months)	HR	Dominant subtype
C1	70	5	7.1	37	50.6	1.596*	Pediatric Low Grade Gliomas
C2	9	6	66.7	30	44.0	0.003	Pediatric High Grade Gliomas
C3	75	8	10.7	36	48.2	1.823	Pediatric Low Grade Gliomas
C4	50	24	48.0	26	39.9	1.125***	Ependymomal Tumor
C5	14	4	28.6	31	33.0	1.125	Craniopharyngioma, Adamantinomatous Type
Sex	-	-	-	-	-	0.000	
Age	-	-	-	-	-	0.000	



Interpreting the Cox output:

- HR > 1 = higher hazard (worse survival) relative to C1 (reference).
- HR < 1 = lower hazard (better survival).

After then running a model with the raw protein values the model had a cordance of 0.05, a model under 0.5 concordance = random chance. Below 0.5 means the model is predicting in the wrong direction patients it thinks have high risk are actually surviving longer.